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DIGITAL SAT MATH • UNIT 0.01

Digital SAT Diagnostic & Adaptive Strategy Drill

PRACTICE WORKSHEET

20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Name: _____ **Date:** _____ **Score:** _____ / 45

Part I — Student-Produced Response Diagnostic (Questions 1–20)

1 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $2(x-6)=3x-5$, what is the value of x?

Answer: _____

2 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} x+6y=-43 \\ -x-y=13 \end{cases}$$

What is the value of y?

Answer: _____

3 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 6x-3y=-6 \\ -6x+5y=-2 \end{cases}$$

What is the value of y?

Answer: _____

4 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $3(x-2)=-3x-12$, what is the value of x?

Answer: _____

5 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $2x+5=-5x+47$, what is the value of x?

Answer: _____

6 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $6(x+1) = -x + 20$, what is the value of x?

Answer: _____

7 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to $|3x - 9| = 15$?

Answer: _____

8 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the positive solution to $|2x - 4| = 12$?

Answer: _____

9 SPR Foundational Linear Equations in One Variable & Systems of Equations

A two-digit number has a tens digit t and a units digit u. The sum of the digits is 11. If the digits are reversed, the new number is 27 greater than the original. What was the original number?

Answer: _____

10 SPR Foundational Linear Equations in One Variable & Systems of Equations

A farmer has chickens and cows. In total, there are 30 heads and 84 legs. How many cows does the farmer have?

Answer: _____

11 SPR Foundational Linear Equations in One Variable & Systems of Equations

A store sells notebooks for $3 each and pens for $1.50 each. Maria spent $18 on a total of 8 items. How many notebooks did she buy?

Answer: _____

12 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to the equation $|2x - 10| = 4$?

Answer: _____

13 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 6x - 3y = 0 \\ 5x - 6y = 28 \end{cases}$$

What is the value of x?

Answer: _____

14 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 3x - y = -6 \\ -7x - 4y = -24 \end{cases}$$

What is the value of x?

Answer: _____

15 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} -4x + 4y = -28 \\ -x - 7y = 1 \end{cases}$$

What is the value of x?

Answer: _____

16 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $3x - 2y = 9$ and $x + 4y = 17$, what is the value of $5x + 6y$?

Answer: _____

17 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $\begin{cases} ax + by = 12 \\ 4a + 3b = 16 \end{cases}$ and $x = 2, y = 3$, what is the value of a?

Answer: _____

18 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $\begin{cases} 3x - 4y = 7 \\ 2x + 3y = 16 \end{cases}$, what is the value of $x^2 - y^2$?

Answer: _____

19 SPR Foundational Linear Equations in One Variable & Systems of Equations

In the system $\begin{cases} mx + 3y = 9 \\ 4x + 6y = 18 \end{cases}$, for what value of m does the system have infinitely many solutions?

Answer: _____

20 SPR Foundational Linear Equations in One Variable & Systems of Equations

A barge travels 36 miles upstream in 4 hours and returns 36 miles downstream in 2 hours. What is the speed of the current in miles per hour?

Answer: _____

Part II — Multiple-Choice Diagnostic Readiness (Questions 21–45)

21 MCQ

If $3x + 14 = 32$, what is the value of x?

A) 46/3
B) 6
C) 18
D) 7

22 MCQ

If $2(x - 8) = 8$, what is the value of x?

A) 0
B) 12
C) 4
D) -4

23 MCQ

A moving company charges a one-time fee of $30 plus $55 per mile. If the total charge was $305, how many miles was the move?

A 6
B 4
C 7
D 5

24 MCQ

If $\frac{x}{3} + 5 = 13$, what is the value of x?

A 27
B 8
C 39
D 24

25 MCQ

If $8x + 12 = 4x + 36$, what is the value of x?

A 8
B 2
C 6
D 24

26 MCQ

Suppose that $5x = 15$, what is the value of $10x$?

A 15
B 30
C 6
D 3

27 MCQ

A taxi company charges a one-time fee of $25 plus $30 per mile. If the total charge was $175, how many miles was the trip?

A 6
B 4
C 5
D 7

28 MCQ

Suppose that $\frac{x}{2} + 3 = 9$, what is the value of x?

A 12
B 6
C 18
D 14

29 MCQ

Suppose that $5x + 14 = 2x + 38$, what is the value of x?

A 24

B 8

C 10

D 24/7

30 MCQ

Assuming that $4(x - 1) = 16$, what is the value of x?

A 4

B 5

C 3

D 15

31 MCQ

A tutoring center charges a one-time fee of $20 plus $60 per session. If the total charge was $260, how many sessions did the student attend?

A 4

B 3

C 6

D 5

32 MCQ

Assuming that $6x + 7 = 3x + 37$, what is the value of x?

A 30

B 12

C 10

D 10/3

33 MCQ

Under the condition that $4x = 8$, what is the value of $8x$?

A 16

B 8

C 4

D 2

34 MCQ

A tutoring center charges a one-time fee of $50 plus $65 per session. If the total charge was $310, how many sessions did the student attend?

A 6

B 4

C 5

D 3

35 MCQ

Under the condition that $6x + 8 = 5x + 13$, what is the value of x?

A 1

B 5/11

C 7

D 5

36 MCQ

Provided that $4(x - 9) = 48$, what is the value of x?

A 39

B 21

C 12

D 3

37 MCQ

A taxi company charges a one-time fee of $30 plus $40 per mile. If the total charge was $150, how many miles was the trip?

A 2

B 5

C 3

D 4

38 MCQ

Which values of x satisfy $-2x + 3 > 7$?

A $x < -2$

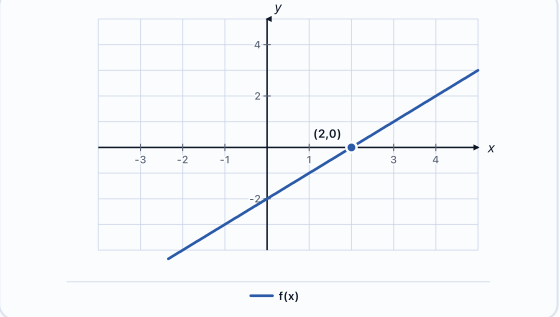
B $x > -2$

C $x > 2$

D $x < 2$

39 MCQ

The graph of a function is shown. At which x-value does the function equal zero?



A $x = 1$

B $x = 2$

C $x = -2$

D $x = 0$

40 MCQ

A line in the xy-plane has a slope of 3 and passes through the point $(2, 11)$. Which equation represents this line?

A $y = 3x + 11$

B $y = 3x - 5$

C $y = 3x + 5$

D $y = 2x + 3$

41 MCQ

Which of the following equations represents a horizontal line passing through the point $(3, -5)$?

A $x = 3$

B $y = 3x - 5$

C $y = -5x + 3$

D $y = -5$

42 MCQ

A gym charges a monthly membership fee of $\$30$ plus $\$2$ for each fitness class attended. If a member attends c fitness classes in a month, which expression represents the member's total cost for that month?

A $30c + 2$

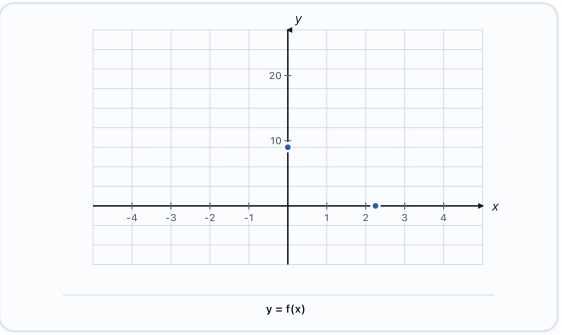
B $2(30 + c)$

C $30 + 2c$

D $\frac{30+c}{2}$

43 MCQ

A line has equation $y = -4x + 9$. What is the y-intercept of the line?



A $(0, 9)$

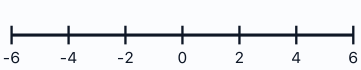
B $(-4, 0)$

C $(0, -4)$

D $(9, 0)$

44 MCQ

Which inequality represents all values of x such that $4x - 5 \leq 19$?



A $x \leq 5$

B $x \leq 7$

C $x \leq 4$

D $x \leq 6$

45 MCQ

What is the slope of the line $2x + 5y = 20$?

A 4

B $\frac{2}{5}$

C $-\frac{2}{5}$

D $-\frac{5}{2}$